Polynomial shears and sign recovery for Diophantine consistency

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Abstract

We construct explicit improvements to the currently supplied Diophantine forms of Con(ZFC). A triangular polynomial change of the final integer coordinate lowers the degree of a ten-integer equation from 1 245 324 to 1 179 788, 1 114 252, or 1 081 125, using respectively 484, 487, and 521 binary arithmetic operations. The predecessor uses 480. The first two substitutions have very short factored formulas; the third attains a lower bound for this particular family of final-coordinate translations. Integral changes of coordinates and monic elimination also give equations in $23, 21, 20, 19, 18$ unrestricted integers at degrees $10, 16, 20, 32, 88$. A separate positive-domain argument gives eleven positive unknowns at degree 211 in 112 operations. All these are fixed ordinary polynomials, with actual coefficient payloads, literal arithmetic data, proofs, and reproducible checks.

A further section develops natural-number selectors and direct proof-checker graphs. These improve an alternative proof-certificate source but do not constitute a completed new global consistency equation. We do not claim a globally simplest equation, a reduction below nine positive unknowns, or a new end-to-end proof-assistant certificate. The completed equations inherit the precisely identified native/compiler/refutation calibration; the new algebraic reductions and their additional sign argument are proved here.

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1 Results, comparison, and meaning of simplicity

The goal is an explicit fixed polynomial P and a specified domain D for which ordinary mathematics formalizable in ZFC proves

$$\mathsf{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in D\ P(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}). \tag{1}$$

Consistency here is the syntactic assertion that there is no finite proof of contradiction in the repository's calibrated recursive ZFC calculus. No choice of coefficient depends on the truth of consistency.

The reviewed checkout is ProveIt commit `c1a102936aca437e9999530a0111e6ddab56dc6d`. It includes the placed sources 31–75 and the five archives then in `docs/incoming`. In particular, the comparisons below include the incoming structural, joint-guard, centering, and radical-descent results; comparing only with the placed wave-ten README would overstate the improvement.

The useful short ten-integer choice is the degree-1 114 252 equation: it saves 131 072 degrees, about 10.53%, for seven extra gates. The degree-1 081 125 choice saves 164 199 degrees, about 13.19%, at a larger arithmetic cost. The eleven-positive equation improves the incoming degree-257, 114-operation construction in both degree and operation count. The new 19-integer equation has degree 32, compared with the incoming 19-integer degree-88 equation; the latter also yields a new 18-integer degree-88 form.

There is no scalar definition of "simplest" inherent in the problem. Number of unknowns, domain, total degree, circuit size, coefficient size, and verification burden are distinct. In particular, an equation in $\mathbb{Z}_{>0}^{11}$ is not an eleven-unrestricted-integer equation. The established nine-positive and

Domain	Unknowns	Exact degree	Additional measurement
$\mathbb{Z}$	10	1 179 788	484 operations
$\mathbb{Z}$	10	1 114 252	487 operations
$\mathbb{Z}$	10	1 081 125	521 operations
$\mathbb{Z}_{>0}$	11	211	112 operations
$\mathbb{Z}$	23	10	11 squared residuals
$\mathbb{Z}$	21	16	9 squared residuals
$\mathbb{Z}$	20	20	8 squared residuals
$\mathbb{Z}$	19	32	7 squared residuals
$\mathbb{Z}$	18	88	6 squared residuals

Table 1: Completed new equations in this delivery. Operations mean live binary $+, -, \times$ gates, including scalar products and squaring; fixed integer leaves and reuse are free. No optimized operation claim is made for the five residual-system equations.

ten-integer variable counts remain relevant. Sun's ten-integer theorem is a recent primary reference for the latter type of compression [2]; Jones's paper is the classical reference for the nine-positive route [1]. Our new changes do not improve those general variable bounds.

2 A general polynomial-shear principle

Lemma 2.1 (Triangular shear)**.** *Let $P \in \mathbb{Z}[\boldsymbol{v}, m]$ and $R \in \mathbb{Z}[\boldsymbol{v}]$. Set*

$$P_R(\boldsymbol{v}, m) = P(\boldsymbol{v}, m + R(\boldsymbol{v})).$$

Then $(\boldsymbol{v}, m) \mapsto (\boldsymbol{v}, m + R(\boldsymbol{v}))$ is a bijection of $\mathbb{Z}^{n+1}$ with inverse $(\boldsymbol{v}, m) \mapsto (\boldsymbol{v}, m - R(\boldsymbol{v}))$. It restricts to a bijection between the integer zeros of P_R and of P.

Proof. Both maps are integral polynomial maps, and their two compositions are the identity. Substituting into the displayed definition of P_R proves the zero-set statement. No positivity, divisibility, or nonvanishing premise is needed. □

The following norm shape makes a suitable shear especially effective:

$$P = \left((\Sigma m - \Theta)^2 - \Sigma^2(a_0 + a_1)\right)^2 - 4\Sigma^4 a_0 a_1. \tag{2}$$

Every displayed symbol other than m is a polynomial in the other coordinates. Suppose R is a polynomial approximation to one radical in the purely algebraic sense that $\varepsilon = a_1 - R^2$ has smaller degree than a_1. Writing $Z_0 = \Sigma m - \Theta$, the shifted equation is exactly

$$P_R = \left(Z_0^2 + 2\Sigma R Z_0 - \Sigma^2(a_0 + \varepsilon)\right)^2 - 4\Sigma^4 a_0 a_1. \tag{3}$$

This follows by expanding $(Z_0 + \Sigma R)^2$ and canceling $R^2 - a_1$. An approximation inequality for square roots is unnecessary. The identity holds on every signed integer assignment, including assignments where a_1 is negative or zero.

Proposition 2.2 (Degree accounting)**.** *Let s, z, r, d_0, d_1, e be upper bounds on the degrees of $\Sigma, Z_0, R, a_0, a_1, \varepsilon$, respectively. Then*

$$\deg P_R \le \max\{2\max(2z, s + r + z, 2s + d_0, 2s + e),\ 4s + d_0 + d_1\}.$$

When one of these contributions strictly dominates all other contributions and its leading homogeneous part is nonzero, the bound is attained.

Proof. Apply $\deg(fg) \le \deg f + \deg g$ and $\deg(f + g) \le \max(\deg f, \deg g)$ to (3). Under strict dominance, lower-degree summands cannot cancel the highest homogeneous part. □

3 The fixed ten-integer source and final norm

3.1 Exact data rather than an unspecified universal parameter

The predecessor is the incoming Round11 H1868 norm-saturated equation [6]. Its literal 480-operation graph is included unchanged as `norm10/upstream/integer10.json`. Its proof, source-packing proof, four native integer payloads, twelve actual live fixed integer leaves, and licenses are retained. The new graphs are complete ordinary polynomial graphs; the predecessor file is used to construct and independently check them, rather than being an existential parameter in the new equations.

The ten unrestricted integer variables are

$$t, g_0, c_0, k_0, \ell, w, x, y, z, m.$$

Use the following source abbreviations:

$$b = 6t + 4, \qquad g = bg_0, \qquad k = b(3k_0 + 2), \qquad B_c = 2^{2134940} b^2.$$

The finite weight list, target list, and three fixed sparse kernel polynomials are the literal data in `norm10/data/code.json` and `norm10/data/kernels.json`. Write them as w_i, q_j, D_0, D_1, D_2, respectively, and set

$$\begin{aligned}
e_c &= \sum_i B_c^{w_i}, & U_c &= e_c + g,\\
G_c &= D_2(B_c)U_c^2 + D_1(B_c)U_c + D_0(B_c), & T_c &= \sum_j B_c^{q_j},\\
N_0 &= B_c^{117}, & N &= B_c^{1986},\\
M_c &= N_0 - 1 - (b-1)e_c, & & \\
S_c &= g + N_0(2G_c + T_c), & T_c' &= M_c + N_0(B_c - 2)T_c - 2,\\
X &= N(S_c + T_c' + 1) - S_c - 1, & & \\
Y &= 2^{34159017} b B_c^{2109}, & \mathcal{C} &= bB_c^{116}.
\end{aligned}$$

These are fixed finite sums and fixed nonnegative powers. Kernel coefficients are explicitly materialized integers, determined by the whole inherited native tuple. Neither a program index chosen by a nonconstructive existence theorem nor a variable exponent occurs in the equation.

3.2 Polynomial registers

Define

$$
\begin{aligned}
&L = \ell Y, \quad U = 2LX, \quad V = 4gwY, \quad W = bw,\\
&h_A = LX(V+1), \quad A = 2h_A, \quad B_1 = 2X+1, \quad C = 6c_0 - 1,\\
&P_h = (LX)UV, \quad K = X + 1 + k(2P_h - 2),\\
&S = 2A - 5, \quad T = 3WC - 2(W^2 - 1),\\
&Q = (P_h^2 - 1)K^2 + 1,\\
&\Delta = 8\mathcal{C}^3 gK^2 - g^2\big(32(C - KL)^2\mathcal{C}^3 + g^2K^2\big),\\
&\xi = zx, \quad D = (A^2 - 4)C^2 + 4,\\
&E_0 = C^2\xi, \quad E = DE_0, \quad E_2 = E_0E,\\
&F = 4(A^2 - 4)E_2 + 1, \quad G = (h_A - 1)(F - 1) - 1,\\
&\Omega = 16E, \quad H = C + F(B_1 - C + \Omega y),\\
&I = (G^2 - 1)H^2 + 1,\\
&\rho = T + Sy, \quad \Sigma = \xi S^5,\\
&\Theta = (\rho^3 - \rho S^2 - S^3)(\rho^2 + 23S^2),\\
&r = bQ, \quad a_0 = r + (9\Delta - 3)(rz)^2, \quad a_1 = DFI.
\end{aligned}
$$

The half-parameter h_A is explicitly the integer polynomial above. No division is being used to define it. Every register is an abbreviation; only the original ten coordinates are quantified.

The supplied predecessor theorem says that the polynomial (2) has an integer zero exactly when the calibrated calculus has a finite refutation. We inherit that theorem, including its Pell and native-source proof obligations. The present section's transformations do not change any of its source parameters or coefficient payloads.

4 Three new ten-integer equations

4.1 Four additional gates

Take

$$R_0 = 2AEGH. \tag{4}$$

The new equation is (2) with m replaced everywhere by $m + R_0$. The essential cancellation is

$$DF - (2AE)^2 = D - 16E^2. \tag{5}$$

Indeed $DF = 4(A^2 - 4)E^2 + D$. Consequently,

$$a_1 - R_0^2 = (D - 16E^2)G^2H^2 + DF(1 - H^2). \tag{6}$$

The shared register $2A$ already occurs in $S = 2A - 5$. Only three new multiplications and the addition to m are needed. The exact graph has 484 live operations.

4.2 Seven additional gates

A better short substitution is

$$R_1 = 2AE_0(D - 2C^2)GH. \tag{7}$$

Put $P_1 = 2AE_0(D - 2C^2)$ and $u = C^2$. Direct polynomial expansion gives

$$DF - P_1^2 = D + 16E_0^2\big(u(3u-4)A^2 - 16(u-1)^2\big). \tag{8}$$

For a transparent derivation, put $A = 2h_A$ and $D = 4D_*$, $D_* = u(h_A^2 - 1) + 1$. Then

$$DF = 256(h_A^2 - 1)u^2 D_*^2 \xi^2 + 4D_*, \qquad P_1 = 8u\xi h_A(2uh_A^2 + 2 - 3u).$$

The h_A^6 and h_A^4 terms cancel on subtraction; the remaining expression is exactly (8). Hence

$$a_1 - R_1^2 = (DF - P_1^2)G^2H^2 + DF(1 - H^2). \tag{9}$$

Compared with (6), two further powers of the high-degree parameter A disappear. The resulting graph has 487 operations.

4.3 A six-term polynomial part

Let $h = h_A$, $u = C^2$, and $J = B_1 - C$. Define the following short polynomial:

$$\begin{aligned} T_* &= \xi J\big(u(2h^2 - 11) + 6\big) \\ &\quad + 16u\xi^2 y\big(u^2(8h^4 - 52h^2 + 143) + u(32h^2 - 176) + 48\big) + 3y, \qquad (10) \\ R_6 &= 32768u^7\xi^4 h^9\big((h-1)T_* - 2y\big). \qquad (11) \end{aligned}$$

This polynomial has integer coefficients. It is obtained by matching the six highest powers $h^{14}, \ldots, h^9$ in a formal square root of a_1. That discovery description is not needed as a correctness assumption: the finite formula (11) and exact expansion of $a_1 - R_6^2$ are the certificate.

For reproducibility, the file `norm10/ROOT_CERTIFICATES.json` gives the full coefficient list of R_6 and of its defect. In the five independent indeterminates (h, B_1, C, ξ, y), R_6 has 26 monomials and the defect has 2306. The independent standard-library checker multiplies the polynomials and checks every coefficient. The defect has degree at most 22 in h; its relevant total-degree bound after source substitution is proved next. The complete literal graph uses 521 live operations.

4.4 Exact total degrees

The source has

$$\deg h_A = \deg A = 16384, \qquad \deg B_1 = 7942, \qquad \deg C = 1, \quad \deg \xi = 2, \quad \deg y = 1.$$

The weighted degrees in these five indeterminates therefore give valid upper bounds after substitution. For the actual source the relevant degrees are

Register	Degree	Register	Degree
D	32770	E	32774
F	65546	G	81930
H	98321	a_1	458818
Σ	81922	Θ	81925
a_0	294619	$\Sigma m - \Theta$	81925

Every proposed R has degree 229409. The cancellation identities give

$$\deg(a_1 - R_0^2) = 426050, \qquad \deg(a_1 - R_1^2) = 393282,$$

and the full finite defect certificate gives

$$\deg(a_1 - R_6^2) \le 368449.$$

Applying Proposition 2.2 yields respectively

$$1\,179\,788, \qquad 1\,114\,252, \qquad 1\,081\,125.$$

These are exact degrees, not merely syntactic upper bounds. For R_0 and R_1, the highest term comes from the square of $-\Sigma^2(a_1 - R^2)$. For R_6, the square in (3) has degree at most $1\,064\,586$, whereas

$$\deg(-4\Sigma^4 a_0 a_1) = 4(81922) + 294619 + 458818 = 1\,081\,125.$$

There can be no cancellation between these unequal degrees.

The checker evaluates the claimed leading homogeneous coefficient after the diagonal specialization of all ten variables at a common parameter, using the actual fixed integers reduced modulo three explicitly supplied primes. For R_6, the residues at $1000003, 1000033, 1000037$ are $347224, 762141, 563833$, respectively. Each nonzero residue proves that the integer leading coefficient is nonzero. A single nonzero residue suffices; the additional residues are transcription controls. The leading terms for the first two variants are likewise certified. This proves attainment of the upper bounds without constructing million-degree expanded polynomials.

Theorem 4.1. *The three literal polynomials in* `norm10/integer10_linear.json`, `norm10/integer10_cubic.json`, *and* `norm10/integer10_sixterm.json` *have the variable counts, exact degrees, and operation counts shown in Table 1. Each is related to the pinned predecessor by an integral triangular polynomial automorphism. Composing with the inherited predecessor theorem gives* (1) *for each of these polynomials on* $\mathbb{Z}^{10}$.

Proof. The literal graphs implement (2) with the respective formulas (4), (7), and (11) substituted for the shift. Lemma 2.1 proves both existence implications, on all signed assignments. The preceding degree calculation and finite graph counts prove the quantitative statements. The source-to-refutation equivalence is exactly the inherited theorem, and negating existence yields the assertion. □

4.5 A lower bound for this restricted transformation family

There is a useful stopping criterion. Keep the entire source, Σ, Θ, a_0, and a_1 fixed, and allow only polynomial translations of m with integer coefficients. In every resulting equation,

$$P_R = \mathcal{B}_R^2 - 4\Sigma^4 a_0 a_1$$

for some integer polynomial $\mathcal{B}_R$. The second summand has the odd degree $1\,081\,125$. The nonzero square $\mathcal{B}_R^2$ has even degree. If its degree is larger, it cannot cancel against the second summand; if it is smaller, it cannot cancel that summand either. Equality of degrees is impossible by parity. The case $\mathcal{B}_R = 0$ is immediate. Thus every such translation has degree at least $1\,081\,125$. The six-term choice attains this bound.

This is an exact optimality statement for a sharply specified transformation family. It is not a lower bound for arbitrary polynomial coordinate changes, alternative norm representations, source encodings, or all Diophantine equations expressing consistency.

5 Fewer unrestricted integer variables

5.1 Double centering and a transport pivot over the integers

The following five equations use unrestricted integer variables and the same six fixed coefficients throughout. Each is a sum of squares of the residuals in its accompanying JSON packet.

Packet	Variables	Residuals	Exact degree
`centered23_degree10`	23	11	10
`pivot21_degree16`	21	9	16
`pivot20_degree20`	20	8	20
`pivot19_degree32`	19	7	32
`pivot18_degree88`	18	6	88

The inspected incoming Joint Guards package had the corresponding points 23/12, 21/20, 20/32, and 19/88. The incoming Radical Guard Descent 22/14 form remains another useful tradeoff. These improvements do not lower the ten-integer high-degree variable bound.

All new coordinate transformations below are integer-polynomial bijections of the indicated parent zero sets. Consequently they preserve the parents' ordinary, ZFC-formalizable equivalence with finite syntactic ZFC refutability. Their unsolvability is therefore ZFC-provably equivalent to Con(ZFC). This is an ordinary mathematical composition; no new end-to-end kernel certificate or object-language ZFC derivation is claimed.

The constituent two-root centering appears in incoming Budget and Centering, and the monic transport pivot appears in incoming Round11. The new result is their application to the newer Joint Guards systems. No historical priority or global optimality claim is made.

5.2 Fixed coefficients

The coherent tuple is

$$\beta = 2^{1064921} - 1, \quad K_0, M_C, M_F, \quad A_* = 2^{1113}, \quad B_* = 2^{666104}.$$

Here K_0, M_C, M_F are exactly the source-34 `mixed-parity-unrounded-1064921` native tuple, with M_F already shifted once. The six actual minimal unsigned little-endian byte strings are supplied, compressed with gzip, with a complete binding manifest. Their combined raw size is 615,883 bytes. These are fixed numeral leaves, not quantified parameters.

5.3 The 23-variable equation

Take independent integer coordinates

$$(J, F, \alpha, z, f, h, r, v, w, \tau, \eta, \zeta, \nu, \gamma, o, j, E, a, I, t, p_1, p_2, p_3).$$

Define polynomial abbreviations only:

$$\begin{aligned}
q &= \beta(3J+1)+1, & W &= 4(3w+1), & X &= qW,\\
Y &= a - E, & k &= \eta + \zeta, & c &= Yk + \eta,\\
V &= fo - c, & C &= q - F - \alpha, & H &= 4a + 3,\\
\Delta &= a^2 + H, & T &= E(c - \eta), & S &= X + \gamma H.
\end{aligned}$$

Also put

$$R = ((q-1)(q-F)+\alpha)(q^2-1) + (M_C + qM_F)(3J+1),$$
$$G = (A_*F)(B_*C) - (q - A_*F - B_*C)^2,$$
$$Z = \eta\zeta, \qquad \mathrm{Pos} = 1 + p_1^2 + p_2^2 + p_3(p_3+1).$$

The eleven residuals are

$$\begin{aligned}
B_1 &= a - E - (2v+1)q^3,\\
B_2 &= E - XY,\\
B_3 &= \tau^2 + (2\tau - k)T - 1,\\
B_4 &= 2acS + S^2 - Hc^2 - 1,\\
B_5 &= k - 1 - hE - R,\\
B_6 &= (K_0 + W)C - F - z(q-1),\\
B_7 &= I - (2r+1)c^2,\\
B_8 &= I^2 - \Delta(f^2-1),\\
B_9 &= (V+\nu)^2 - I^2\nu(2V+\nu) - 1,\\
B_{10} &= V + k - 1 - cj - hE,\\
B_{11} &= GZ - (t^2 - G - Z + 1)^2 - \mathrm{Pos}.
\end{aligned}$$

Thus the complete ordinary polynomial equation is

$$\boxed{P_{23} = \sum_{i=1}^{11} B_i^2 = 0.}$$

Every displayed abbreviation is inlined in the supplied packet; it introduces neither a variable nor a side condition.

For the proof, the incoming 25-variable octic has independent c, V, y and uses $k_0 = 1 - V + jc + hE$ in its c equation and first norm. It retains the row $L = k - k_0$. Replace k_0 by k using polynomial multiples of this retained row. Then make the triangular root translations

$$\tau_{\mathrm{old}} = \tau + E(c - \eta), \qquad y_{\mathrm{old}} = V + \nu,$$

and eliminate the two monic definitions $c = (a-E)k + \eta$ and $V = fo - c$. The three cancellation identities are

$$(\tau+T)^2 - T(k+T) - 1 = \tau^2 + (2\tau - k)T - 1,$$
$$(ac+S)^2 - (a^2+H)c^2 - 1 = 2acS + S^2 - Hc^2 - 1,$$
$$I^2(V^2 - (V+\nu)^2) + (V+\nu)^2 - 1 = (V+\nu)^2 - I^2\nu(2V+\nu) - 1.$$

In the literal parent order, its thirteen restored rows are exactly

$$B_1, B_2, (a-E)B_{10}, B_3 + TB_{10}, B_4, B_5 - B_{10}, B_6, 0, B_7, B_8, B_9, B_{10}, B_{11}.$$

This gives both directions of the zero-set equivalence without any division. The inverse root translations and coordinate projection are unique. The residual degrees are

$$4, 3, 4, 5, 4, 2, 5, 4, 5, 3, 4.$$

Only B_7 contains r, and

$$[r^2 a^4 \eta^4] P_{23} = 4.$$

Hence the total degree is exactly ten on the actual fixed-coefficient slice.

5.4 A transport pivot that costs no dominant degree

All four remaining parents contain

$$(K_0 + W)(q - F - \alpha) - F - z(q - 1) = 0.$$

Change the complement coordinate by $C = q - F - \alpha$, then solve the now monic row for F:

$$F = (K_0 + W)C - z(q - 1), \qquad \alpha = q - F - C.$$

Delete that row and F, and substitute everywhere else. Both the map and its inverse use only integer polynomials, so this is a bijection of integer zero sets. The packing register becomes

$$R = (q - F)(q^3 - q) - C(q^2 - 1) + (M_C + qM_F)(3J + 1),$$

of degree at most five. The determinant G has degree four, and its contribution to the joint-sign residual has degree at most eight. The larger Pell/sign residuals therefore already pay the resulting degree cost in the selected parents. This yields the 21/16, 20/20, 19/32, and 18/88 packets in the table.

5.5 The complete 18-variable endpoint

Take independent integer coordinates

$$(J, C, z, f, h, r, v, w, \tau, y, \gamma, o, j, c, t, p_1, p_2, p_3).$$

In addition to q, W, X and the transport pivot above, define

$$\begin{aligned}
Y &= (2v + 1)q^3, & E &= XY, & a &= (X + 1)Y,\\
V &= fo - c, & k &= 1 - V + jc + hE, & \eta &= c - Yk,\\
\zeta &= k - \eta, & H &= 4a + 3, & \Delta &= a^2 + H,\\
I &= (2r + 1)c^2, & T &= E(c - \eta), & D &= X + ac + \gamma H.
\end{aligned}$$

Use the same R, G, Z, Pos definitions. The six residuals are

$$\begin{aligned}
A_1 &= \tau^2 - T(k + T) - 1,\\
A_2 &= D^2 - \Delta c^2 - 1,\\
A_3 &= cj - V - R,\\
A_4 &= I^2 - \Delta(f^2 - 1),\\
A_5 &= I^2(V^2 - y^2) + y^2 - 1,\\
A_6 &= GZ - (t^2 - G - Z + 1)^2 - \mathrm{Pos}.
\end{aligned}$$

The equation is

$$\boxed{P_{18} = \sum_{i=1}^{6} A_i^2 = 0.}$$

These six residuals are completely inlined in its packet, so the count 18 includes all quantified variables. Retaining η as a nineteenth independent coordinate and adding $c - \eta - Yk = 0$ gives the degree-32 sibling.

5.6 Degree certificates and verification

The four pivot equations have the following exact nonzero coefficients:

$$\begin{aligned}[a^4f^8o^4]P_{21} &= 1,\\ [f^4v^4J^{12}]P_{20} &= 2^4 3^{12}\beta^{12},\\ [\eta^4h^4w^4v^4J^{16}]P_{19} &= 2^{12}3^{20}\beta^{16},\\ [h^8w^8v^{16}J^{56}]P_{18} &= 2^{32}3^{64}\beta^{56}.\end{aligned}$$

For the last two, the leading forms are

$$E_{\mathrm{top}} = 1944\beta^4J^4wv, \qquad \eta_{\mathrm{top}}\big|_{P_{18}} = -104976\beta^7hJ^7wv^2.$$

In P_{19}, the dominant joint-sign part is $-(\eta hE_{\mathrm{top}})^2$ before the outer SOS square. In P_{18}, it is $-\eta_{\mathrm{top}}^4$. These are the unique contributions at the respective highest residual degrees. The first two certificates come from unchanged reduced-auxiliary and strong-Pell rows; the pivoted guard cannot contribute their selected monomials. Full derivations are given in the companion proof.

The generator authenticates declaration closure, exact degrees and all leading coefficients. A separate checker imports no builder: it checks all thirteen centering identities, every literal pivot identity, the actual six coefficient byte strings and additional signed modular transcription diagnostics. The finite tests certify only the algebra and the data stated; the universal native/Pell/refutation calibration remains an explicitly inherited ordinary theorem.

The next concrete targets are a positivity certificate that avoids squaring the degree-eight sign register in P_{19}, and an orientation theorem that avoids the degree-eleven η substitution in P_{18}. Either would improve these equations more than further cosmetic factorization.

6 An eleven-positive equation of degree 211

6.1 Eleven positive unknowns: an auxiliary offset and two root centers lower degree to 211

The following construction has eleven strictly positive integer unknowns, exact degree 211, and 112 live binary arithmetic operations. The centered predecessor in the incoming joint-guards package has degree 257 in 114 operations; the uncentered Pell-descent version has degree 277 in 111 operations. The improvement is therefore a strict improvement in both degree and evaluator size over the centered predecessor.

The new step is the affine substitution

$$T_{\mathrm{old}} = T + F + \alpha.$$

It makes the auxiliary quantity positive at an earlier stage of the proof, so an expensive exceptional-sign term can be removed from the transport weight. A second center absorbs the strong root, leaving a degree-two gap bound. This is an existence-preserving substitution on canonical source witnesses, not a bijection of the entire positive zero sets.

6.2 The complete polynomial

Use the coherent compact44 fixed tuple (β, K_0, M_C, M_F), where $\beta = 2^{590069} - 1$ and M_F is already shifted. The tuple is defined by the retained finite recipe and actual integer payloads. The unknowns are

$$j, F, \alpha, i, T, X, Y, n, \eta, \zeta, \gamma > 0.$$

Every remaining symbol is a polynomial abbreviation. Define

$$\begin{aligned}
J &= 3j+1, \qquad q = \beta J + 1, \qquad C = q - F - \alpha,\\
E &= XY, \qquad k = \eta + \zeta, \qquad c = kY + \eta,\\
a &= E + Y, \qquad H = 4a + 3, \qquad \Delta = a^2 + H = (a+2)^2 - 1,\\
\ell &= EkY, \qquad I = ic^2, \qquad M = q^3(q-1),\\
R &= ((q-1)(q-F)+\alpha)(q^2-1) + (M_C + qM_F)J,\\
B_t &= (qK_0 + X)C - qF, \qquad D_\tau = \ell(\ell+k)+1,\\
D_f &= 1 + \Delta I^2, \qquad Q = \Delta^2 I^2, \qquad \delta = Q - 1,\\
T_{\text{old}} &= T + F + \alpha, \qquad C_f = (a+2)I,\\
V &= D_f(cT_{\text{old}} - R) - c, \qquad L = kT_{\text{old}},\\
W &= 1 + (q-1)L,\\
\overline{Z} &= M(\ell + k + n + L + V + C_f) + (q-1)Y - q^3 W B_t,\\
r_0 &= k - R - 1, \qquad S = 4qY, \qquad U = qS,\\
D_* &= U(X + ac) + H(U\gamma + SX - r_0),\\
A_1 &= D_*^2 - U^2(1 + \Delta c^2).
\end{aligned}$$

The three-root norm is

$$\begin{aligned}
A &= M^2 D_\tau, \qquad B_f = M^2 D_f, \qquad C_{\text{num}} = M^2(QV^2 - 1),\\
T_N &= \delta^2 A B_f, \qquad R_N = \delta C_{\text{num}},\\
J_N &= \delta(\overline{Z}^2 - A - B_f) - C_{\text{num}},\\
N_c &= J_N^2 - 4[T_N + R_N(A + B_f)],\\
N_8 &= N_c^2 - 64 T_N R_N \overline{Z}^2.
\end{aligned}$$

The proposed single polynomial equation is

$$\boxed{P_{11} = N_8 + A_1 + N_8 A_1 = 0.} \tag{12}$$

Its literal graph is `p11_positive_degree211.json`.

6.3 Recovering radicals before any transport sign

Put, only in this proof,

$$t = \sqrt{D_\tau}, \qquad f = \sqrt{D_f}, \qquad y = \sqrt{\frac{QV^2 - 1}{Q - 1}}.$$

The unconditional three-root separation argument of source70 continues to apply: it requires only $T_{\text{old}} > 0$ and $I \geq c^2$, not positivity of C. We recall its substance. Let $b = cT_{\text{old}} - R$. If $b = 0$, then $V = -c$, $y < c + 1$, and $f - t > (a+1)c(c-1) > c + 1$. If $b \neq 0$, then

$$y > |V| \geq D_f - c > 2f > f + t.$$

Here $f > 3c^2$ and $f > t$ follow directly from $\Delta \geq 15$ and $c > kY$. Thus one of the three roots strictly dominates the other two. Their positive rational radicands are pairwise distinct: $D_\tau < D_f < Q$,

and equality of the third radicand to either integer $D \in \{D_\tau, D_f\}$ would imply $Q(D - V^2) = D - 1$, impossible since $0 < D - 1 < Q$.

A rational signed sum of three such roots forces them all rational. For signed roots x_1, x_2, x_3 with rational nonzero sum s and squares a, b, c, set

$$u = \frac{s^2 - a - b - c}{2}, \qquad v = \frac{u^2 - ab - ac - bc}{2s}.$$

Then

$$x_1 = \frac{a(a+u)}{as+v}, \qquad as + v = x_1(x_1 + x_2)(x_1 + x_3) \neq 0,$$

and the cyclic formulas prove the assertion.

The universal norm identity

$$\left[(x^2 - a - b - c)^2 - 4(ab + ac + bc)\right]^2 - 64abcx^2 = \prod_{\varepsilon \in \{-1,1\}^3} (x - \varepsilon_1\sqrt{a} - \varepsilon_2\sqrt{b} - \varepsilon_3\sqrt{c})$$

therefore shows that $N_8 = 0$ supplies rational signed roots with

$$\frac{\overline{Z}}{M} = \tau_* + f_* + y_*, \qquad \tau_*^2 = D_\tau, \quad f_*^2 = D_f, \quad QV^2 - (Q-1)y_*^2 = 1.$$

The first two roots are integers. The adjacent-coefficient Pell descent of source69 makes y_* an integer as well. In that descent, writing $y_* = a/m$ in lowest terms gives $m^2 \mid Q - 1$. After replacing V and y_* by their absolute values at each step, the transformation

$$(V, y_*) \longmapsto ((2Q-1)V - 2(Q-1)y_*,\ (2Q-1)y_* - 2QV)$$

preserves the norm and the denominator, while strictly reducing $|V| > 1$. Strong induction reaches $|V| = 1$, where $|y_*| = 1$.

Since $M \mid \overline{Z}$, reduction modulo q^3 and $q - 1$ now gives

$$q^3 \mid Y, \qquad q - 1 \mid B_t.$$

No sign of B_t has been used. The equation $A_1 = 0$ also gives an integral main root D_*/U, whose absolute value can be classified. Writing χ_A, ψ_A for the usual Pell sequences, positive classification supplies indices $m_1, p \geq 1$ such that

$$|\tau_*| = \chi_{2XY^2+1}(m_1), \qquad k = 2\psi_{2XY^2+1}(m_1), \qquad c = \psi_{a+2}(p).$$

Because $c \geq 3$, $p \geq 2$. The case $m_1 = 1$ would give $k = 2$, $\eta = \zeta = 1$, $c = 2Y + 1$, contradicting $c \geq 2(a+2) \geq 4Y + 4$. Hence

$$c > kY \geq 4(2XY^2 + 1)Y > 8XY^3 \geq 8q^9. \tag{13}$$

This is established before $C > 0$, before the source bounds on R, and before interpreting any recovered transport gap.

6.4 The affine offset eliminates the exceptional-sign case

Define the proof-only bound $\widehat{R} = q^3(q + F + \alpha)$. The fixed-mask estimates give unconditionally

$$|R| < \widehat{R}.$$

Put $h = F + \alpha \geq 2$. Since $q \geq 16$,

$$q^3(q + h) \leq q^3(q/2 + 1)h < q^4 h < ch$$

by (13). Thus

$$cT_{\text{old}} - R = cT + ch - R > cT \geq c, \qquad V > D_f c - c > 1.$$

The new auxiliary coordinate is therefore positive at the required point of the proof. This is what permits deletion of the predecessor's $4qD_f$ exceptional-sign weight.

The same bound $ch > \widehat{R}$ also gives

$$0 < cT_{\text{old}} - R < c(T + 2h), \qquad 0 < V < D_f c(T + 2h).$$

Moreover,

$$\delta - (\Delta - 1)D_f = \Delta(I^2 - 1) > 0, \qquad c < k(Y + 1), \qquad \Delta - 1 > 2(Y + 1).$$

The last inequality follows from $a \geq 2Y$ and $Y \geq 1$. Therefore

$$0 < y - V = \frac{V^2 - 1}{\delta(y + V)} < \frac{V}{\delta} < \frac{c(T + 2h)}{\Delta - 1} < \frac{k(T + 2h)}{2} < k(T + h) = L.$$

Also $f < C_f = (a + 2)I$, since $C_f^2 - D_f = I^2 - 1 > 0$. We conclude

$$L + C_f + V - f - y > 0. \tag{14}$$

Let $s = Y/q^3 > 0$ and $b = B_t/(q - 1) \in \mathbb{Z}$. The recovered norm identity reads

$$\tau_* + f_* + y_* = \ell + k + n + L + V + C_f + s - Wb.$$

If $b \leq 0$, its right side is strictly larger than $|\tau_*| + |f_*| + |y_*|$, because $|\tau_*| < \ell + k$ and (14) holds. Hence $B_t > 0$, which forces $C > 0$. Only now do we invoke the native source bounds $3q < R < q^4 - 2$.

Set

$$f_{\text{old}} = |f_*|, \quad i_{\text{old}} = i, \quad u = |y_*| - V, \quad r = n + L + V + C_f - f_* - y_*.$$

All are positive integers, with $r > n$. They satisfy the weighted source70 carrier with weight $h_* = L$, its normalized strong norm, its auxiliary norm, and its adaptive main norm. That inherited interface is sound for every positive integer h_*. Its native/compiler and syntactic-refutation calibration therefore supplies a finite ZFC refutation.

6.5 Completeness and the exact cost of the offset

A finite refutation gives the canonical compact source, with $R \equiv 3 \pmod 4$, $R > 3q$, $F + \alpha < q$, and the earlier bound on c. Choose its normalized strong pair f, I, $I = ic^2$, and write $M_* = \Delta I$. The inherited positive auxiliary reconstruction uses

$$V_* = \frac{\chi_{M_*}(R)}{M_*}, \qquad y_* = \psi_{M_*}(R), \qquad T_{\text{old}} = \frac{V_* + c + Rf^2}{cf^2}.$$

It proves the last quantity integral and $y_* > V_* > 0$. The stronger bound needed for the new offset is elementary. Since $R \geq 7$,

$$V_* \geq M_*^{R-1} \geq M_*^6, \qquad f^2 < M_*^2, \qquad M_* \geq c^2,$$

and hence

$$T_{\mathrm{old}} > \frac{V_*}{cf^2} > \frac{M_*^4}{c} \geq c^7 > q > F + \alpha.$$

Thus $T = T_{\mathrm{old}} - F - \alpha > 0$ is integral. No new cofinality assumption or congruence on the strong Pell pair has been introduced.

The same growth also proves $T_{\mathrm{old}} > C_f$: because $I \geq c^2$,

$$T_{\mathrm{old}} > \frac{M_*^4}{c} = \Delta^4 I \frac{I^3}{c} > (a+2)I = C_f.$$

After these choices compute L, W. We have $L = kT_{\mathrm{old}} > C_f$ and $L > k$. The canonical transport has $t_0 = B_t/(q-1) \geq q$, $s < k$, and $\tau > \ell$, so

$$r = Wt_0 + \tau - \ell - k - s > qW - 2k > 2L.$$

Consequently

$$n = r - L - V_* - C_f + f + y_* = r - L - C_f + f + (y_* - V_*) > 0.$$

This makes $\overline{Z} = M(\tau + f + y_*)$ and $N_8 = 0$; the inherited adaptive carrier supplies positive γ and $A_1 = 0$. Completeness is proved on canonical witnesses, which is exactly what an existence equivalence requires.

Both residuals are squares modulo four. Thus the factorization

$$1 + P_{11} = (1 + N_8)(1 + A_1)$$

shows $P_{11} = 0$ if and only if $N_8 = A_1 = 0$: a negative unit would require a residual -2, impossible modulo four.

6.6 Exact degree and verification

The degree audit uses an exact universal double-center identity. For independent variables z, a, b, c, e, f, set

$$x = z + b + c, \quad B = b^2 - e, \quad C = c^2 - f, \quad N_c^\circ = (x^2 - a - B - C)^2 - 4(aB + aC + BC),$$

$$N^\circ = (N_c^\circ)^2 - 64aBCx^2.$$

Give (z, a, b, c, e, f) the weights $(8, 16, 11, 22, 18, 26)$. Exact expansion, recorded in a 229-term certificate, gives

$$[N_c^\circ]_{63} = 8bc^2 z, \qquad [N^\circ]_{126} = 64b^2c^4(z^2 - a).$$

Substitute

$$z = \overline{Z} - MV - MC_f, \quad a = A, \quad b = MC_f, \quad c = MV,$$

$$e = M^2(I^2 - 1), \qquad f = -\frac{M^2(V^2 - 1)}{\delta}.$$

These are exact because $C_f^2 - D_f = I^2 - 1$ and $(MV)^2 - C_{\mathrm{num}}/\delta = -M^2(V^2-1)/\delta$. Multiplication by δ^2 and δ^4 gives the actual inner and outer norm. The power of f is at most four, so the cleared identity is polynomial. Since $\deg \delta = 18$, the exact residual degrees are

$$\deg N_c = 99, \qquad \deg N_8 = 198, \qquad \deg A_1 = 13, \qquad \deg P_{11} = 211.$$

For an attained-degree certificate on the actual fixed tuple, put

$$\begin{aligned} q_0 &= 3\beta j, \qquad k_0 = \eta + \zeta, \qquad I_0 = ik_0^2 Y^2, \\ D_{f,0} &= X^2Y^2I_0^2, \qquad Q_0 = X^4Y^4I_0^2, \qquad C_{f,0} = XYI_0, \\ L_0 &= k_0(T + F + \alpha), \qquad \ell_0 = Xk_0Y^2, \qquad R_0 = q_0^3(q_0 - F), \\ B_{t,0} &= (K_0q_0 + X)(q_0 - F - \alpha) - q_0F. \end{aligned}$$

Then

$$\begin{aligned} [N_8]_{198} &= 64Q_0^4q_0^{32}D_{f,0}^4R_0^4C_{f,0}^2L_0B_{t,0}(L_0B_{t,0} - 2\ell_0), \\ [A_1]_{13} &= 32q_0^2k_0X^2Y^4(4q_0^2Y\gamma + R_0). \end{aligned}$$

The coefficient of

$$j^{48}i^{18}\eta^{39}X^{30}Y^{67}T^2F^4\alpha^2\gamma$$

is $2^{13}(3\beta)^{48} > 0$, independently of K_0, M_C, M_F. The literal graph has 64 multiplications, 31 additions, and 17 subtractions. Scalar multiplication and squaring are charged; fixed numeral leaves and register reuse are free. The two-residual system has 109 operations, and the single equation has 112.

The separate exact audit checks every named polynomial contract, the exact double-center cancellation, the eight-conjugate identity, radical-extraction identities, the attained leading coefficient, liveness, and operation counts. The complete compact44 payload is verified and independently reconstructed from its finite recipe. Normal and optimized Python runs pass with explicit checks.

Combining soundness and completeness with the named inherited source proves the ordinary mathematical consistency equivalence on the stated positive domain. It does not lower the existing nine-positive variable count, prove global minimality, or provide a new end-to-end formal calibration. The new content is the offset, the resulting early sign argument, its canonical completeness proof, and the degree-211 endpoint.

7 Alternative direct proof sources

7.1 Optimal natural selector weights

This section gives a completed local improvement to the direct commutative-certificate architecture. Its global bounded closure has not been compiled, so the local dimensions below are not a new final consistency-equation variable count.

Theorem 7.1 (Natural interpolation weights)**.** *Let $k \geq 1$, and let $G_0, \ldots, G_{k-1}$ be integer polynomials nonnegative on natural inputs. Set*

$$g_k(r) = \prod_{j=0}^{k-1}(r - j), \qquad L_i(r) = \prod_{\substack{0 \leq j < k \\ j \neq i}} (r - j),$$

and define

$$W_i(r) = \begin{cases} L_i(r), & k-1-i \text{ even}, \\ (r-k)L_i(r), & k-1-i \text{ odd}. \end{cases}$$

Then g_k and every W_i are nonnegative on $\mathbb{N}$, and

$$\exists r \in \mathbb{N}: \quad g_k(r) + \sum_{i=0}^{k-1} W_i(r) G_i = 0 \quad \Longleftrightarrow \quad \bigvee_{i=0}^{k-1} G_i = 0.$$

Each W_i has the smallest possible degree among real polynomials nonnegative on $\mathbb{N}$, positive at i, and zero at every other integer in $\{0, \dots, k-1\}$: degree $k-1$ in the even case and k in the odd case.

Proof. For $r < k$, the factors give $g_k(r) = 0$ and $W_i(r) = 0$ unless $r = i$. At $r = i$, the sign of $L_i(i)$ is $(-1)^{k-1-i}$; the optional factor $i - k$ changes the negative sign. For $r \geq k$, all relevant factors are nonnegative, and $g_k(r) > 0$. Thus every summand in the displayed equation is nonnegative, a zero forces $r < k$, and its unique surviving branch weight is positive. This proves both directions.

For optimality, $k-1$ required distinct roots force degree at least $k-1$. At that degree a candidate equals cL_i; nonnegativity at large natural arguments forces $c > 0$. Its value at i has the required sign exactly in the even case. The odd case therefore needs degree at least k, attained by the stated construction. □

The incoming Round11 source uses twelve nonnegative Horn branch polynomials $h_0, \dots, h_{11}$, with respective degrees 2, ten copies of 4, and 8 for the bundle branch h_9. The truth branch is $h_0 = p^2 + q^2 + v^2$. Retain $M = M_0 + 1$ and the accumulator residuals

$$\begin{aligned} E &= S'(1 + vM) - S(1 + pM)(1 + qM), \\ F &= R' - R((1+p)(1+q) + v). \end{aligned}$$

At a truth step $p = q = v = 0$, the equations say $S' = S$ and $R' = R$. All such steps can therefore be deleted from a finite certificate, preserving its endpoints and final height. The existential trace length and its finite CRT code are replaced by those of the shortened trace. The distinguished endpoints are $S_0 = M + 1$ and $S_N = 1$, with $M > 0$, so a successful trace cannot consist entirely of deleted steps. Conversely every trace without these steps is already an old trace. This establishes equality of the global existential source predicates; it does not assert equality of their local zero sets at a fixed step.

Group the remaining eleven factors as

$$(G_0, \dots, G_5) = (h_9, h_1h_2, h_3h_4, h_5h_6, h_7h_8, h_{10}h_{11}).$$

Every group has degree eight. The resulting literal local polynomial is

$$\mathcal{L} = g_6(r) + \sum_{i=0}^{5} W_i(r) G_i + E^2 + F^2.$$

Its degree is exactly 14, with 21 local natural witnesses and five exposed state/parameter coordinates. The former three-bit construction uses 23 local witnesses at the same degree. Exact sparse expansion of the supplied rows gives 50 461 monomials; the leading term of $\mathcal{L}(t, \dots, t)$ is $723072t^{14}$. The source JSON, selector weights, entire expanded polynomial, and a standard-library verifier are included. The exact degree assertion is therefore independently reproducible.

There is a second improvement using two ternary selectors u, w. The $k = 3$ weights are

$$U_0(x) = (x-1)(x-2), \quad U_1(x) = x(x-2)(x-3), \quad U_2(x) = x(x-1),$$

of degrees $2, 3, 2$. Use the group array

$$(G_{ij}) = \begin{pmatrix} h_9 & h_5 & h_1h_2 \\ h_6 & h_7 & h_8 \\ h_3h_4 & h_{10} & h_{11} \end{pmatrix}$$

and the polynomial

$$\mathcal{L}_{12} = g_3(u) + g_3(w) + \sum_{i,j=0}^{2} U_i(u)U_j(w)G_{ij} + E^2 + F^2.$$

The degree-eight groups occupy corners with weight degree four. Every other group contributes at degree at most ten. Thus this source has 22 local witnesses and degree exactly 12; its 40 990-monomial expansion has leading coefficient 627840 under uniform substitution. Nonnegativity and the two selector penalties prove the same local and global source equivalence. The new local tradeoffs are consequently $(21, 14)$ and $(22, 12)$ in place of $(23, 14)$.

The incoming index-free inert-prime closure has conditional coordinate count $2m + 9 + 2f + p + (m + 1)b$, for local witness dimension m and chosen factorial, power, and binomial graph costs f, p, b. Holding the remaining graph conventions fixed, the change $m = 23 \longrightarrow 21$ saves $4 + 2b$ coordinates at that interface. This is a conditional compiler accounting identity; it does not replace materialization of the entire direct consistency polynomial. The degree-twelve alternative instead saves $2 + b$ coordinates under that same convention.

7.2 Exact local arithmetic graphs

A second improvement exploits the natural domain directly. For natural $x_1, \dots, x_s$, their sum vanishes exactly when they all vanish. Hence a mask $(\sum x_i)F^2$ can replace $(\sum x_i^2)F^2$ without changing its zeros. In particular,

$$C(n, p) = (n + p)(n - p - 1)^2$$

is a nonnegative cubic with zeros exactly at $p = \mathrm{pred}(n)$. The corresponding signed quadratic residual $(n + p)(n - p - 1)$ has the same zero set and may be squared when a strict residual-system compiler is required.

More significantly, the quadratic polynomial

$$Z(n, b, p) = (n + b - p - 1)^2 + bp$$

jointly represents predecessor and zero test:

$$Z = 0 \quad \Longleftrightarrow \quad p = \mathrm{pred}(n), \qquad b = \mathbf{1}_{n=0}.$$

Indeed $bp = 0$ and $n + b = p + 1$. If $b = 0$, then $n = p + 1 \geq 1$; if $p = 0$, then $n + b = 1$, giving only $(n, b) = (0, 1)$ or $(1, 0)$. The converse is immediate. Hiding p gives a one-witness zero-test graph of degree two. If both outputs are already present, no extra witness is needed.

For a conditional, one may use the one-witness quartic

$$n(c - b)^2 + \big((c - a)(n - s - 1)\big)^2,$$

or the two-witness cubic

$$Z(n, t, p) + t(c - a)^2 + n(c - b)^2.$$

Their zeros project exactly to $c = a$ when $n = 0$, and $c = b$ otherwise. For the quartic, $n = 0$ makes $n - s - 1$ nonzero for every natural s; $n > 0$ forces the first branch directly. For the cubic, the joint graph fixes the exact branch bit. Completeness uses the canonical predecessor and bit.

Keeping the incoming two-witness max-square pairing polynomial Q unchanged, the tree projections use $Q(p, a, b; t, s) + C(n, p)$. The actual found-marker list head uses

$$Q(p, a, b; t, s) + (n + p + c)(n - p - 1)^2 + n(c - a - 1)^2.$$

At $n = 0$ the middle term forces $p = c = 0$; at $n > 0$ it forces $p = n - 1$ and the last term forces $c = a + 1$. Thus nil and the stored-zero marker are both preserved. All naturalness assumptions are explicit.

Local graph	Hidden witnesses	Degree	Tree operations
Predecessor	0	$4 \to 3$	$10 \to 8$
Zero test	1	$4 \to 2$	$14 \to 10$
Tree projection/list tail	4	4	$49 \to 47$
Conditional	1	4	$16 \to 15$
List head	5	4	$59 \to 55$

The comparison is against the incoming Joint Guards package and counts literal binary-expression trees, including repeated square operands and the same final addition of zero. Full expansions and separate shared-DAG counts are supplied. These are exact local graph improvements, without a claimed end-to-end compiler or object-level formal certificate.

8 Logical calibration and what is verified

All completed equations have literal finite arithmetic descriptions with fixed integer coefficients. The ten-integer group retains the Round11 H1868 tuple; the integer residual group retains the coherent six-numeral joint-guard tuple; the positive group retains the complete compact44 tuple. These tuples must not be mixed. A symbol that appears in one group need not denote the same integer in another.

The completed reductions prove the following chain:

$$\text{new zero exists} \iff \text{calibrated predecessor zero exists} \iff \text{finite ZFC refutation exists.}$$

For the polynomial shears and monic eliminations, the first equivalence is an explicit zero-set bijection. The positive-domain component additionally uses the proved sign-recovery and canonical-witness estimates. The second equivalence imports the inherited native/compiler/calculus theorem. Its ordinary ZFC calibration is a stated mathematical theorem, not a conclusion drawn from an external Lean theorem. The pinned primary K215 proof [4] explicitly establishes

$$\mathsf{ZFC} \vdash \big(\mathrm{Halt}(K215, \mathrm{blank}) \longleftrightarrow \neg\, \mathrm{Con}(\mathsf{ZFC})\big).$$

It composes the dominated-calculus refutation theorem, a quantified source-to-machine simulation, and the exact finite blank startup. The native arithmetic representation and cited Pell/compiler proofs then connect this fixed halting predicate with the predecessor equations. Our integral polynomial transformations and sign argument compose with those ordinary proofs within ZFC, giving (1).

The distinction from the remaining object-calculus work is explained in the retained source-63 scope review [5]. The ordinary argument codes finite runs and derivations in ambient ZFC and uses uniform induction over its entire ω. Its model-theoretic steps use set models, their satisfaction relations, and first-order completeness; they do not assume transitive models or external well-foundedness.

Thus the claim uses the identified ordinary mathematical calibration as a dependency. It does not promote correctness on external numerals to correctness on every internal natural number. No appeal to the actual truth of Con(ZFC) is made at any stage.

There are three separate verification claims:

1. The new universal algebraic and arithmetic implications are proved in this manuscript, with exact coefficient certificates where an expansion is too long to print.

2. The supplied programs check literal polynomial transcription, coefficient payloads, graph topology, exact algebra, and degree witnesses. Finite signed evaluations are secondary regression controls; they are not proofs of an unbounded semantic equivalence.

3. A fresh end-to-end Lean/Rocq proof, or an explicit sealed object-language ZFC derivation, is not supplied. The repository's `OBJECT_ZFC.md` correctly distinguishes an external equivalence from an internal derivation. This delivery does not close those formalization obligations by renaming them.

The direct-proof source in Section 7 has a different boundary: its local source theorem is proved, but it is not substituted for the completed native source without constructing and measuring the full bounded-quantifier elimination. It therefore does not support a claimed new global number of variables or polynomial degree.

9 Reproduction and integration

`README.md` gives commands and the file map. Run the root `verify.py` from any working directory. Standard-library checkers are provided for the ten-integer and positive components. The integer elimination and independent symbolic audits document their SymPy requirements explicitly; the alternative generators need only the standard library. All acceptance checks use explicit exceptions, rather than depending on Python's removable `assert` statements. The root check records which commands actually ran and whether they passed.

The archive contains the editable manuscript, compiled PDF, complete new literal equations, all required actual coefficient payloads, source pins, the relevant predecessor interfaces, independent audits, licenses, and notices. The optional article build uses pdfLaTeX and the packages named in this file. It uses no network connection. The source snapshot and incoming archive identities are fixed in `PROVENANCE.json`.

For repository intake, place the archive's wrapper directory in a new proposal subdirectory under `Computability/DiophantineConsistency/con-zfc-diophantine-forms`. No directory number is reserved here. The delivery makes no edit to the repository's main branch or existing packages. Comparisons should retain both domains and all measured tradeoffs, particularly the operation cost of the six-term ten-integer variant.

10 Further research questions

1. **Go beyond translations of the last coordinate.** The odd-degree argument proves that $1\,081\,125$ is the floor for polynomial translations of m with the present four norm inputs fixed. Can a joint triangular change of y, z, or a source parameter lower $\deg(\Sigma^4 a_0 a_1)$ without increasing the circuit cost excessively? This question has a clear obstruction and a concrete target.

2. **Reduce the cost of the six-term shear.** Its factored expression adds 41 live operations. Can the 26-monomial polynomial share additional powers and products with D, F, G, H and

the source prefix? A claimed improvement should be accompanied by a literal graph and exact identity, rather than a heuristic cost estimate.

3. **Audit other radical norms for removable leading squares.** The method applies whenever a large radicand has an integral polynomial part. For unrestricted coordinates, completeness is automatic under a triangular shear. Positive coordinates require a new range argument; a formal square-root expansion by itself cannot supply that argument.

4. **Improve the degree-211 positive endpoint.** The new transport offset supplies a strict sign before native code recovery. Can a smaller positive offset do the same uniformly? Can the majorant or root center be simplified while preserving the order of the sign-recovery argument? A negative or zero intermediate must not be excluded by an unstated positivity assumption.

5. **Continue the monic elimination search with degree control.** The 18-integer form has six residuals and degree 88. Search for other polynomial changes that expose a monic row while keeping the largest residual below degree 44. A Gröbner-style symbolic search can assist, but the final certificate should give explicit integral restoration maps.

6. **Compile the direct certificate source and measure it.** The natural-domain selector and functional graph improvements reduce local cost without requiring K215. The next necessary object is a complete ordinary-polynomial constructor with its actual witness, gate, and coefficient counts, including bounded-universal elimination. Local counts alone are not a size comparison with the finished equations.

7. **Close the internal-ZFC proof boundary.** Formalize the actual coefficient-bound native compiler, its source calibration, and the new simple shear/monic-elimination lemmas in the repository's sealed object-language framework. The new local algebra is short; uniform correctness of the inherited coding over the internal natural numbers of every ZFC model remains the substantial work.

References

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[3] V. Reshetnikov, *ProveIt*, inspected repository snapshot `c1a102936aca437e9999530a0111e6ddab56dc6d`. https://github.com/VladimirReshetnikov/ProveIt. See the DiophantineConsistency catalogue and its source-pinned research packages; their individual notices are retained with reused artifacts.

[4] ProveIt contributors, *Ordinary proof of the combined215 machine*, `Computability/BusyBeaver/Research/ZFC/candidate-k215/proof.md`, at the inspected repository snapshot. Retained verbatim as `calibration/K215_CALIBRATION_PROOF.md`; its Git blob is `5b62cecff23a798a4b34255d1f843e72ad11b623`.

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[6] ProveIt incoming package, *Structural reductions for Diophantine consistency equations, Round eleven*, archive `ConZFC_Structural_Reductions_Round11_2026-10-24.zip`, especially `lucas_h1868/PROOF.md`. Exact archive and member pins are recorded in the accompanying provenance files.

[7] ProveIt incoming package, *Joint sign guards and Pell descent for smaller consistency equations*, `ZFC_Diophantine_Joint_Guards_2026-10-24.zip`.

[8] ProveIt incoming package, *Joint guards and two Pell-root translations*, `con-zfc-budget-and-centering-2026-10-23.zip`.

[9] ProveIt incoming package, *Radical majorants, guard compression, and Pell descent*, `con-zfc-radical-guard-descent-2026-10-23.zip`.

[10] ProveIt source 68, *Joint positivity and free Lucas coordinates*, `68-joint-positivity`, notably its direct certificate source.

[11] ProveIt source 71, *Calibrated determinant guards*, `71-determinant-guards`.

[12] ProveIt source 75, *Signed-power recovery and inert-prime certificates for ZFC consistency*, `75-signed-bootstrap`.